

Adaptive sensor management for UGV monitoring based on risk maps

Seoyeon Kim^a, Young-Hoon Jung^b, Hong Min^c, Taesik Kim^{d,*}, Jinman Jung^{e,*}

^a Institute of Human-Centered Computing, Inha University, 100, Inha-ro, Michuhol-gu, Incheon, 22212, Republic of Korea

^b Department of Civil Engineering, Kyung Hee University, 1732 Deogyong-daero, Giheung-gu, Yongin-si, 17104, Gyeonggi-do, Republic of Korea

^c School of Computing, Gachon University, 1342 Seongnam-daero, Sujeong-gu, Seongnam, 13120, Gyeonggi-do, Republic of Korea

^d Department of Civil and Environmental Engineering, Hongik University, 94 Wausan-ro, Mapo-gu, Seoul, 04066, Republic of Korea

^e Department of Computer Engineering, Inha University, 100, Inha-ro, Michuhol-gu, Incheon, 22212, Republic of Korea

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ABSTRACT

Because of the recent advances in sensor technology, unmanned ground vehicles (UGVs) are equipped with various high-performance sensors to improve their mission performance. The energy consumption of the sensors is a critical issue because most UGVs operate on rechargeable batteries while conducting their missions. In particular, efficient sensing has become a crucial challenge due to the growing demand for advanced intelligent monitoring missions. However, while previous studies have focused on power consumption models for vehicle locomotion, they have not fully addressed the high energy consumption of sensors.

This study proposes an adaptive sensor-management algorithm to address the critical issue of energy consumption by considering the monitoring and navigation sensors that are widely used in UGV monitoring. After characterizing the proposed adaptive sensor-management strategy based on a risk map composed of monitoring and saving zones, we present a power consumption model to quantify the energy savings of the sensors. Furthermore, the optimal interval was derived to minimize power consumption during UGV monitoring. The proposed algorithm can minimize the energy consumption by adjusting the optimal interval according to the mission environment. We demonstrate the results of our analysis through various evaluations.

1. Introduction

In recent years, unmanned ground vehicles (UGVs) have become increasingly popular owing to their ability to perform various tasks without human intervention [1]. UGVs have evolved into intelligent systems that incorporate high-performance sensors to perform functions such as environmental perception, localization, navigation, and path planning [2]. Efficient energy consumption is crucial because UGVs rely on rechargeable electric batteries with limited energy capacity [3]. Many studies have analyzed energy consumption by focusing on vehicle locomotion [4–10]. These power consumption models can play a crucial role in predicting the energy consumption of UGV movements during mission execution.

With the growing demand for autonomous vehicles and artificial intelligence (AI), UGVs have become equipped with high-performance computing resources and sensors that significantly increase their power consumption demands [11–13]. The typical energy consumption of these sensors and processors can range from a few watts to hundreds of watts, depending on specific hardware parameters, such as the sampling rate [14]. To conduct a comprehensive analysis of the

energy consumption of UGVs, it is important to focus not only on the energy consumed during vehicle locomotion but also on the energy consumption of the sensors in the electrical/electronic system.

Furthermore, energy consumption is an important issue to consider in the field of autonomous UGV monitoring with limited power. Numerous studies have been conducted to minimize power consumption in the data collection and processing of UGV sensors [14–28]. Several approaches have been focused on minimizing the sensor energy consumption through adaptive sensor management or path planning according to the monitoring environment. Risk maps (e.g., ground settlement risk, fire risk, and gas risk) are commonly used in environmental monitoring to provide a comprehensive view of the risk level within the mission area [26–31]. The risk map can help the UGV improve the safety or efficiency of the mission in real time, based on its current location and associated risk level. One approach when using a risk map is avoiding potential high-risk areas on the risk map to ensure the safety of the mission. Another approach is to increase monitoring efficiency by selectively monitoring or prioritizing high-risk areas rather than scanning the entire mission area. For example,

* Corresponding authors.

E-mail addresses: skim@inha.ac.kr (S. Kim), jyounghoon@khu.ac.kr (Y.-H. Jung), hmin@gachon.ac.kr (H. Min), taesik.kim@hongik.ac.kr (T. Kim), jmjung@inha.ac.kr (J. Jung).

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when passing through zones with high risk levels on the risk map, the UGV may need to perform intensive monitoring to observe potential hazards by increasing the sampling rate of the monitoring sensor. Conversely, when passing through zones with low risk levels, the UGV can adopt energy-saving strategies. In this approach, accurate detection of whether the UGV is in a high- or low-risk zone is critical for achieving energy-efficient monitoring. Over-monitoring in zones where it is unnecessary can waste energy, whereas energy-saving strategies in zones where monitoring is essential can potentially lead to mission failure.

However, existing methods assume, based on idealized assumptions of its detection capabilities, that a UGV can accurately detect and distinguish between areas where monitoring is required and those where monitoring is unnecessary. In practical scenarios, there may be discrepancies between the detected and actual zone states owing to the detection interval of navigation sensors, such as the inertial navigation and global navigation satellite systems. Increasing the sensing frequency of navigation sensors can improve the accuracy of zone detection, but it also increases energy consumption. Therefore, it is crucial to balance the tradeoff between energy consumption and accurate zone detection.

In this study, we propose an adaptive sensor-management strategy for UGV monitoring based on a risk map, which represents each zone with one of two simple risk states: a monitoring zone (M) with relatively high risk that requires intensive monitoring, and a saving zone (S) with relatively low risk that allows for energy-saving strategies. We first investigate the power consumption model of our adaptive approach based on the risk map by considering the monitoring sensor (MS) and navigation sensor (NS). To address practical energy consumption concerns, we do not assume an ideal scenario in which the NS can accurately detect the current zone state as M or S. Instead, by considering the occurrence of false positives or false negatives in zone detection, we derive an optimal detection interval based on the distribution of zone states on the risk map. Finally, we propose an adaptive sensor-management algorithm that uses the optimal detection interval. This allows the UGV system to adaptively maintain the balance of power consumption in a variety of monitoring environments. The results of this analysis provide a basis for improving the energy efficiency of various autonomous vehicles.

The remainder of this paper is organized as follows. In Section 2, we present recent studies on power consumption models for UGVs and energy-efficient UGV monitoring. In Section 3, we introduce our adaptive sensor-management strategy and present a new power consumption model for the adaptive approach based on a risk map. After analyzing the optimal power consumption, we improved our adaptive sensor-management algorithm for UGV monitoring. In Section 4, we evaluate the performance of our adaptive sensor-management algorithm through various tests. Finally, conclusions are drawn and potential directions for future work are presented in Section 5.

2. Related works

2.1. Energy consumption of UGVs

Considerable research effort has been dedicated to improving the efficiency of UGVs during missions. Most studies focused on vehicle locomotion, which is the main factor affecting the energy consumption of UGVs. In [4], a prediction model for the power consumption of UGVs was proposed. This model, referred to as the vehicle dynamic model, considers various factors, such as road grade, road surface conditions, and driving style, to provide a more accurate prediction of power consumption. This study provides a better understanding of the energy consumption of UGVs during movement. In [5], an extension study was conducted to predict the adaptive mission energy requirements through an energy model to prevent mission failure owing to energy

consumption for remotely controlled UGVs. The authors improved energy prediction using a recursive least squares estimation and Bayesian prediction method based on the availability of prior mission knowledge. Other studies have been conducted to determine the most energy-efficient path by considering obstacles [6] and factors such as rolling resistance and path length [7]. In various fields of application, such as robotics [8–10], predicting the optimal energy consumption continues to be heavily researched.

2.2. Energy-efficient UGV monitoring

Many existing studies have focused on reducing the energy consumption in the mechanical movements of UGVs. However, recent UGVs are required to meet higher demands for automation and AI, which require the use of high-performance hardware for data collection, navigation, and processing [11,12]. The increased energy consumption of various sensors, along with the energy consumption associated with traditional locomotion, has become a key challenge for UGVs.

Numerous studies have been conducted to minimize the power consumption in the data collection, navigation, and processing of UGV sensors. Most of these studies aimed to minimize power consumption by employing dynamic sampling strategies that focus on collecting only the necessary data while reducing duplicity based on the domain context. This is similar to the conventional dynamic methods in WSN research that adjust the sleep mode or sampling rate according to the detected environment (e.g., perceived level of risk [15], traffic [16], and mobility [17]). In recent UGV research [18], various approaches have been employed to reduce energy consumption, such as using the sleep mode or controlling the horizontal angular resolution when the vehicle is stationary or when mission objects are not detected at a long distance. Moreover, in [19], the authors used an adaptive method of adjusting the sensing rate according to the dynamics of the physical environment to minimize the energy consumption of autonomous vehicles.

Furthermore, there is research focusing on communication energy in UGVs, aiming to find the best trade-off between moving energy and communicating energy [20,21]. [20] introduces an EL-UGV model that combines UGV and Edge Learning to address IoT communication energy constraints. The model finds the path to solve the path loss in wireless communication and utilizes a sample size planning module to optimize learning performance. In [21], a model combining UGV and backscatter communication is proposed to enhance IoT data communication quality in wireless networks. It optimizes UGV movement based on noise levels, aiming for energy-efficient data communication.

In autonomous UGV applications, the use of large deep-learning models for environmental perception is prevalent. The optimization of data processing for deep-learning models through lightweight research can also reduce the computational energy consumption of UGVs. Trade-offs between accuracy and energy consumption in deep neural networks have been analyzed to optimize and lighten the networks for autonomous vehicles [22,23]. In [24], a new neural network optimization method called knowledge-based pruning was used for autonomous driving applications. Furthermore, fusion sensor research is being conducted to optimize the processing efficiency by utilizing customized sensors through a combination of multiple sensors that are specifically tailored to match the processing needs of the application in different environments. In [25], a sensor fusion system was proposed that integrates a thermal infrared camera and a light detection and ranging (LiDAR) sensor that can be controlled according to the visibility conditions, such as during the day or night. The authors in [14] presented an energy-aware sensor fusion approach that adjusts the fusion method based on context to minimize energy consumption while maintaining perception performance.

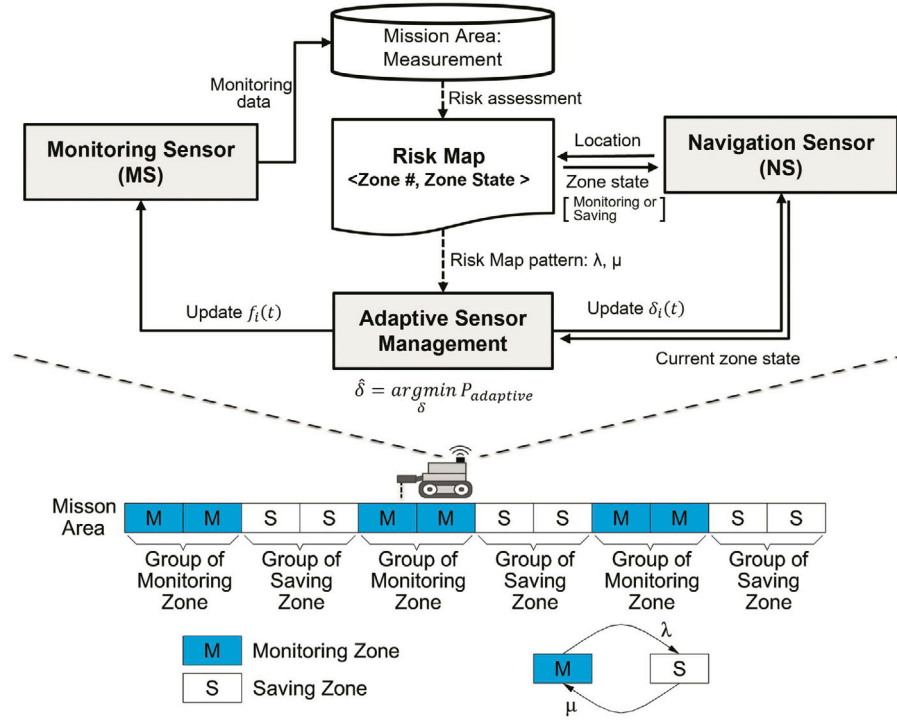


Fig. 1. UGV with a monitoring mission based on a risk map.

2.3. Energy-efficient UGV monitoring based on risk maps

Recently, the use of monitoring sensors incorporated in UGVs, such as high-resolution cameras, 3D laser profiler sensors, and LiDAR sensors, has varied depending on the monitoring mission. Moreover, UGVs are commonly equipped with navigation sensors, such as inertial navigation and global navigation satellite systems, for navigation and localization in conjunction with environmental maps of the mission area. For the efficient monitoring of autonomous vehicles, the use of risk maps to represent the level of risk in different regions has been widely adopted. In [26], the entire mission region was divided into safe zones and red zones to minimize the time and energy consumption by avoiding excessive scanning of the entire mission area. Another study calculated the fire risk of forest fields across an entire map and conducted research on efficient path determination [27]. The authors of [28] proposed an adaptive sampling strategy based on Gaussian process models that focused on the most valuable and informative regions for improved measurements.

Existing studies on energy-efficient sensor management based on risk maps have significantly reduced energy consumption using dynamic sampling and path planning optimization. However, little research has been conducted on precise energy consumption analysis. Our approach to UGV monitoring based on a risk map differs from those of previous studies in several ways. First, we conducted a precise analysis of the energy consumption, providing a comprehensive understanding of the power consumption for UGV monitoring based on the risk map. In contrast to previous studies, our analysis does not rely on the ideal assumption of accurate zone detection by NS; instead, it determines the optimal zone detection interval by considering the occurrence of false positives or negatives during zone detection. Furthermore, our approach considers the overall power consumption of both the MS and NS, which distinguishes it from previous studies that focused only on a single sensor.

3. Power consumption model for energy-efficient UGV sensing

3.1. System model

We consider a UGV denoted by v_i , with a monitoring mission. The mission area is divided into zone units, as shown in Fig. 1. Thus, we consider a zone-based risk map, where each zone is assigned a specific risk value from two zone states: M, which requires intensive monitoring, and S, which does not require intensive monitoring. We assume that the zone states on the risk map are provided as prior information through a domain-specific risk assessment. To analyze the risk distribution in the mission area, we define a zone group as a sequence of consecutive zones with the same state, either all M states or all S states. As the UGV traverses the zones in the mission area over time, it alternates between the M and S groups. The cycles of M and S groups can be represented as state transition rates in a stochastic process that alternates between two states, known as the alternating renewal process [32]. Specifically, let λ denote the transition rate from an M state to S state, and let μ denote the transition rate from an S state to M state. Assuming that the UGV moves through the zones in the mission area at a constant speed, we can calculate λ and μ as the inverse of the expected time that the UGV spends in M and S states, respectively. The transition rates of λ and μ are dependent on the mobility pattern of the UGV and the risk map patterns of M and S in the mission area. In the example of the mission area illustrated in Fig. 1, assuming that the UGV has a mobile pattern that moves through one zone per time unit at a constant speed, the UGV remains in each state for an average of two time units. According to the definition of λ and μ as the inverse of the expected time spent in each state, $\lambda = 0.5$ and $\mu = 0.5$.

To explain our risk map-based UGV monitoring, we started with two fundamental sensors: an NS and MS. The NS is responsible for detecting the zone state (M or S) based on the current location from the risk map, which enables our adaptive sensor-management controller to make decisions about whether to intensively monitor the zone, whereas the MS is responsible for collecting data into a database for measurement.

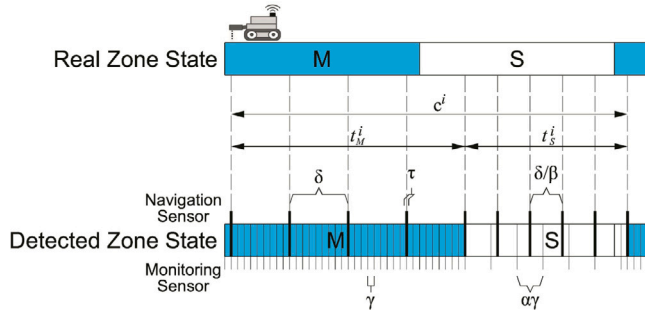


Fig. 2. Real and detected zone states by the UGV system.

Table 1

Notation used in the paper.

Notation	Description
V	The set of UGVs. $V = \{v_1, v_2, \dots, v_N\}$
$s_i(t)$	The zone state of whether v_i is in the saving zone (S) or in the monitoring zone (M) detected by NS according to the current location at time t , i.e., $s_i(t) \in \{M, S\}$
$f_i(t)$	The sampling rate of MS of v_i at time t
$\delta_i(t)$	The detection interval of NS of v_i to identify the zone state at time t
α	The reduction ratio of the sampling rate of MS sensor in S compared with M
β	The reduction ratio of the detection interval of NS sensor in S compared with M
λ	The transition rate from M to S
μ	The transition rate from S to M
P_M	The expected power consumption in M
P_S	The expected power consumption in S
P_ϵ	The expected power consumption required to perform a zone detection by NS
τ	The time taken to detect once
$E[t_M]$	The expected time in M
$E[t_S]$	The expected time in S
$E[X_M]$	The expected number of detections in M before first S state detection
$E[X_S]$	The expected number of detections in S before first M state detection

By conducting a risk assessment of the measurements collected by the MS in each zone, the risk map can be updated online or offline. The mission is considered successful when the MS collects data that meet the monitoring requirements in the entire mission area. We assume that a relatively low sampling rate is required in S for less intensive monitoring.

Specifically, we reduce the MS sampling rate in S by a factor of α . By doing so, we can significantly reduce energy consumption while still meeting the mission requirements. To achieve this objective, it is essential for the NS to accurately detect S, where the MS can reduce power consumption. If the detection interval of NS is too short, it can accurately detect the zones but also increases energy consumption. Conversely, if it is too long, the power consumption can also increase because of the failure to find saving zones. In particular, if the transition from S to M is unsuccessful, it can lead to mission failure. Therefore, determining the optimal detection interval is crucial to reduce power consumption. We use a factor of β to shorten the detection interval of NS in S. In summary, our adaptive sensor-management system uses a reduced sampling rate of MS by a factor of α and a shortened detection interval of NS by a factor of β in S compared with M.

Fig. 2 illustrates how the proposed system dynamically controls the MS and NS in two different zones. The zone states detected by the UGV may differ from the real zone states. We assume that both the

sampling rate of MS, denoted by $f_i(t)$, and the detection interval of NS, denoted by $\delta_i(t)$, can be dynamically adjusted in the system. Let f_s denote the default sampling frequency of MS that is the inverse of sampling interval γ . As shown in Fig. 2, NS periodically detects whether the UGV is in M or S based on the current location every δ time interval. MS adaptively decreases $f_i(t)$ by the ratio of α when the current location is S for the energy-saving as follows.

$$f_i(t) = \begin{cases} f_s & \text{if } s_i(t) = M \\ \frac{1}{\alpha} \cdot f_s & \text{else if } s_i(t) = S \end{cases} \quad (1)$$

Here, $s_i(t)$ is the zone state of whether v_i is in M or in S detected by NS according to the current location at time t . By contrast, NS detects more frequently in S by shortening the interval with a ratio of β for the accurate mission.

$$\delta_i(t) = \begin{cases} \delta & \text{if } s_i(t) = M \\ \frac{1}{\beta} \cdot \delta & \text{else if } s_i(t) = S \end{cases} \quad (2)$$

The goal of our proposed system is to optimize power consumption by changing the sampling rate of the MS and the detection interval of the NS according to the zone state. Table 1 lists the notation used in this paper.

3.2. Analysis of power consumption

First, we analyzed the expected power consumption of our adaptive sensor-management strategy. We focused on the power consumed by MS and NS. In the simplest model, the power consumption per unit time is proportional to the sampling frequency of the sensor [33]. The power consumption of the MS differs in M and S, as shown in Eq. (3).

$$P_M = P_0 \cdot f_s, \quad P_S = P_0 \cdot \frac{f_s}{\alpha} \quad (3)$$

Under this assumption, NS also consumes more energy with an increase in the number of detection attempts.

The zone state observed by the UGV appears to be alternately repeated between the M and S states. Let t_M and t_S denote the time required to continuously observe the M and S states (as independent random variables), respectively. The expected power consumption can be expressed by Eq. (4).

$$P_{\text{adaptive}} = \frac{P_M \cdot E[t_M] + P_S \cdot E[t_S] + P_\epsilon \cdot \tau \cdot (E[X_M] + E[X_S])}{E[t_M] + E[t_S]} \quad (4)$$

where τ is the time taken to detect once, P_ϵ is the power consumption required to detect once by NS, and X_M and X_S are random variables that model the number of detection attempts during t_M and t_S , respectively.

In probability and statistics, the geometric distribution describes the probability of the first success occurring after k number of trials. Each detection attempt of NS consists of independent trials with only two possible outcomes, and the success probability Pr_S in Eq. (6) remains the same for each detection attempt. Hence, we assume that random variable X_M follows a geometric distribution, allowing us to calculate the number of attempts until the first detection of the S state, as shown in Eq. (5).

$$Pr(X_M = n) = (1 - Pr_S)^{n-1} \cdot Pr_S \quad (5)$$

Here, the probability of transition from the M state to the S state by NS, denoted by Pr_S , can be calculated using the transient state probability.

$$Pr_S = P\{s(t + \delta) = S | s(t) = M\} = \frac{\lambda}{\lambda + \mu} \left(1 - e^{-(\lambda + \mu)\delta}\right) \quad (6)$$

Since the mean of a geometric distribution $E[X_M]$ is $\frac{1}{Pr_S}$, $E[t_M]$ can be expressed as a function of δ as shown in Eq. (7); $E[t_M]$ is the time

to detection of M state by NS, which can be expressed as a multiple of δ .

$$\begin{aligned} E[t_M] &= \delta \cdot E[X_M] \\ &= \delta \cdot \frac{1}{Pr_S} \\ &= \frac{\delta \cdot (\lambda + \mu)}{\lambda \cdot (1 - e^{-(\lambda + \mu)\delta})} \end{aligned} \quad (7)$$

Similarly, we can assume that random variable X_S follows a geometric distribution in a similar manner. The main difference is that the success probability for random variable X_S , is defined as Pr_M in Eq. (8), distinguishing it from random variable X_M . The probability of a transition from the S to the M state by NS, Pr_M , can be expressed by Eq. (8).

$$Pr_M = P\left\{s\left(t + \frac{\delta}{\beta}\right) = M | s(t) = S\right\} = \frac{\mu}{\lambda + \mu} \left(1 - e^{-(\lambda + \mu)\frac{\delta}{\beta}}\right) \quad (8)$$

Therefore, $E[t_S]$ can also be calculated using Eq. (9). Recall that NS periodically detects with a time interval of $\frac{\delta}{\beta}$ seconds in the S state as shown in Eq. (2).

$$\begin{aligned} E[t_S] &= \frac{\delta}{\beta} \cdot E[X_S] \\ &= \frac{\delta}{\beta} \cdot \frac{1}{Pr_M} \\ &= \frac{\delta \cdot (\lambda + \mu)}{\beta \cdot \mu \left(1 - e^{-(\lambda + \mu)\frac{\delta}{\beta}}\right)} \end{aligned} \quad (9)$$

By substituting Eqs. (7) and (9) into Eq. (4), the expected power consumption can be calculated using Eq. (10). This power consumption model reflects both the sampling rate of the MS and the detection interval of the NS.

$$\begin{aligned} P_{adaptive} &= \frac{P_M \cdot E[t_M] + P_S \cdot E[t_S] + P_\epsilon \cdot \tau \cdot (E[X_M] + E[X_S])}{E[t_M] + E[t_S]} \\ &= \frac{P_M \cdot \delta \cdot \frac{1}{Pr_S} + P_S \cdot \frac{\delta}{\beta} \cdot \frac{1}{Pr_M} + P_\epsilon \cdot \tau \cdot \left(\frac{1}{Pr_S} + \frac{1}{Pr_M}\right)}{\delta \cdot \frac{1}{Pr_S} + \frac{\delta}{\beta} \cdot \frac{1}{Pr_M}} \\ &= \frac{P_M \cdot \beta \cdot \delta \cdot Pr_M + P_S \cdot \delta \cdot Pr_S + P_\epsilon \cdot \beta \tau \cdot (Pr_M + Pr_S)}{\beta \delta \cdot Pr_M + \delta \cdot Pr_S} \end{aligned} \quad (10)$$

3.3. Optimizing the detection interval

One of the main objectives of our study was to investigate the optimal interval $\hat{\delta}$. Fig. 3 shows the effect of the NS detection interval. If the detection interval of the NS is too long, the power consumption increases owing to the failure to find S where the expected power consumption can be decreased. If the actual S in the mission area is not detected (false negative detection of S), unnecessary monitoring may occur, leading to an increase in energy consumption. In addition, a false positive of S can lead to mission failure. In addition, if the interval is too short, the power consumption unnecessarily increases because of the power consumption caused by frequent detections. This indicates that power consumption can be optimally minimized to conduct sensing by detecting S with an optimal NS interval.

We can derive the optimal interval $\hat{\delta}$ from the optimization problem in Eq. (11).

$$\hat{\delta} = \underset{\delta}{\operatorname{argmin}} P_{adaptive} \quad (11)$$

Eq. (10) is differentiated relative to δ and is set to zero. We also used a Taylor series expansion to solve the optimization problem, as

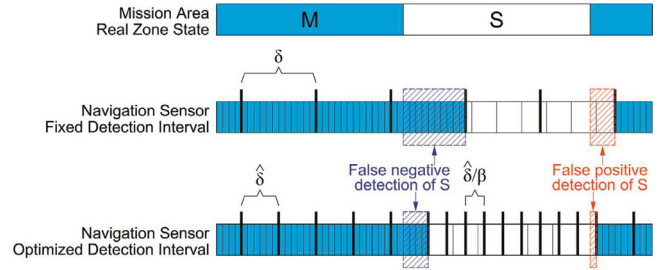


Fig. 3. Effect of the optimal interval of NS.

$\lambda=1.000, \mu=0.111$	M	S	S	S	S	S	S	S	S
$\lambda=0.500, \mu=0.125$	M	M	S	S	S	S	S	S	S
$\lambda=0.333, \mu=0.143$	M	M	M	S	S	S	S	S	S
$\lambda=0.250, \mu=0.167$	M	M	M	M	S	S	S	S	S
$\lambda=0.200, \mu=0.200$	M	M	M	M	M	S	S	S	S
$\lambda=0.167, \mu=0.250$	M	M	M	M	M	M	S	S	S
$\lambda=0.143, \mu=0.333$	M	M	M	M	M	M	M	S	S
$\lambda=0.125, \mu=0.500$	M	M	M	M	M	M	M	M	S
$\lambda=0.111, \mu=1.000$	M	M	M	M	M	M	M	M	M

Fig. 4. Investigated cases for the evaluation of number of monitoring and saving zones.

shown in Eq. (12) for $(\lambda + \mu)\delta < 1$.

$$\begin{aligned} 1 - e^{-(\lambda + \mu)\delta} &= (\lambda + \mu)\delta - \frac{((\lambda + \mu)\delta)^2}{2!} + \frac{((\lambda + \mu)\delta)^3}{3!} \dots \\ 1 - e^{-(\lambda + \mu)\frac{\delta}{\beta}} &= \frac{(\lambda + \mu)\delta}{\beta} - \frac{((\lambda + \mu)\frac{\delta}{\beta})^2}{2!} + \frac{((\lambda + \mu)\frac{\delta}{\beta})^3}{3!} \dots \end{aligned} \quad (12)$$

The differential equations can be approximated using Eq. (13) by considering terms that are less than the second order from Eq. (12). If $(\lambda + \mu)\delta$ is less than 1, then terms of higher than second order are relatively negligible.

$$\frac{dP_{adaptive}}{d\delta} = \lambda\mu(P_M - P_S) \cdot (\beta - 1) \cdot \delta^2 - P_\epsilon \cdot \tau \cdot (\beta - 1)^2 \delta - P_\epsilon \cdot \tau \cdot \beta(\lambda + \mu) \quad (13)$$

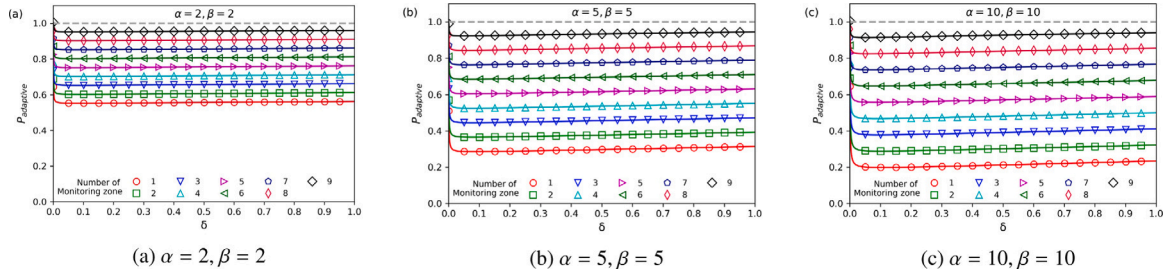
Thus, the optimal interval $\hat{\delta}$ such that Eq. (10) has the minimum power consumption can be easily calculated by finding the root of quadratic Eq. (13), as expressed in Eq. (14).

$$\begin{aligned} \hat{\delta}_{Taylor} &= \\ &= \frac{P_\epsilon \tau \alpha (\beta - 1)^2 + \sqrt{P_\epsilon^2 \tau^2 \alpha^2 (\beta - 1)^4 + 4 P_\epsilon \tau \lambda \mu P_0 f_s \alpha \beta (\alpha - 1) (\beta - 1) (\beta \lambda + \mu)}}{2 \lambda \mu P_0 f_s (\alpha - 1) (\beta - 1)} \end{aligned} \quad (14)$$

Our optimizer uses this equation to show the optimal energy consumption.

3.4. Adaptive sensor management using the optimal detection interval

In this section, we present the proposed adaptive sensor-management algorithm that uses the optimal detection interval $\hat{\delta}$ to achieve minimum power consumption. The proposed algorithm takes the environmental parameters (α , β , λ , and μ) as input values. Our algorithm first initializes the f_s by considering the hardware specification of the MS and the requirements for data collection. It also initializes the NS by applying the optimal detection interval, as shown in Eq. (11). It is triggered whenever the zone state is changed according to the

Fig. 5. $P_{adaptive}$ against δ .

$\lambda=0.200, \mu=0.200$	M	M	M	M	M	S	S	S	S	S
$\lambda=0.400, \mu=0.400$	M	M	M	S	S	M	M	S	S	S
$\lambda=0.600, \mu=0.600$	M	M	S	M	M	S	S	M	S	S
$\lambda=0.800, \mu=0.800$	M	M	S	M	S	M	S	M	S	S
$\lambda=1.000, \mu=1.000$	M	S	M	S	M	S	M	S	M	S

Fig. 6. Investigated cases for the evaluation of number of groups of monitoring and saving zones.

current location of the NS and adaptively adjusts the sampling rate and detection interval according to the current zone state. Especially, our adaptive sensor-management algorithm can dynamically recalculate the optimal interval by considering the updated risk map pattern (λ and μ). Algorithm 1 describes the adaptive sensor-management algorithm using the optimal detection interval $\hat{\delta}$ based on Eq. (11) in detail.

Algorithm 1 Adaptive Sensor-Management Algorithm Using the Optimal Detection Interval

Inputs:given any positive $\alpha, \beta, \lambda, \mu$ **Outputs:**

The sampling rate of MS and detection interval of NS

initialize the default sampling rate according to the specification**calculate the optimal detection interval** $\hat{\delta}$ using Eq. (11)**while** $s_i(t) \neq s_i(t-1)$ every $\delta_i(t)$ time **do****recalculate** $\hat{\delta}$ using Eq. (11)**if** $s_i(t) = S$ **then** $f_i(t) \leftarrow \frac{1}{\alpha} \cdot f_s$ for MS $\delta_i(t) \leftarrow \frac{1}{\beta} \cdot \hat{\delta}$ for NS**else** $f_i(t) \leftarrow f_s$ for MS $\delta_i(t) \leftarrow \hat{\delta}$ for NS**end if****update** λ, μ **end while**

The performance of the proposed algorithm can be compared with those of the fixed sampling and ideal sampling algorithms. In the fixed sampling algorithm, the MS always samples at a fixed rate of f_s without considering the existence of a saving zone, as shown in Eq. (15).

$$P_{fixed} = P_M = P_0 \cdot f_s \quad (15)$$

As the UGV moves through the mission area, it observes events where it alternates between two states (M or S state). The exponential distribution is used to describe the time interval between events assuming that the events can occur randomly. If the UGV detect events ideally, the $E[t_M]$ and $E[t_S]$ in each state can be calculated using the mean of the exponential distribution, as shown in Eq. (16), we can express the power consumption of the ideal sampling algorithm as

$$P_{ideal} \cdot$$

$$E[t|s(t) = M] = \int_0^\infty t \lambda e^{-\lambda t} = \lambda^{-1} \quad (16)$$

$$E[t|s(t) = S] = \int_0^\infty t \mu e^{-\mu t} = \mu^{-1}$$

As shown in Eq. (17), P_{ideal} is the power consumption of the ideal sampling of the MS without the additional power consumption cost caused by the NS.

$$\begin{aligned} P_{ideal} &= \frac{P_M \cdot E[t|s(t) = M] + P_S \cdot E[t|s(t) = S]}{E[t|s(t) = M] + E[t|s(t) = S]} \\ &= \frac{P_M \cdot \lambda^{-1} + P_S \cdot \mu^{-1}}{\lambda^{-1} + \mu^{-1}} \end{aligned} \quad (17)$$

Therefore, the power consumption of the proposed algorithm lies between P_{fixed} and P_{ideal} as the upper and lower bounds, respectively.

4. Performance evaluation

The proposed algorithm was evaluated in two scenarios considering different factors such as the number of monitoring and saving zones and the number of groups of each zone. For a simple investigation, the expected power consumption P_M and P_e were set to be 1 and 0.001, respectively. The value of τ was set to 0.05, and parameters α and β were set as 2, 5, and 10, consecutively.

4.1. Various numbers of monitoring and saving zones

The performance of the proposed algorithm was evaluated as the number of monitoring zones was increased from 1 to 9, as shown in Fig. 4. With a total of 10 zones, the number of saving zones decreased from 9 to 1 as the number of monitoring zones increased. As the number of groups was set to 1, the inverse of the number of monitoring zones and saving zones indicate λ and μ , respectively. For example, when the number of monitoring and saving zone is 1 and 9, the values of λ and μ are 1.000 and 0.111, respectively.

The sensitivity of the proposed algorithm to the sampling interval was limited because it primarily focused on the number of groups of each zone rather than the number of zones. Given a set pair of α and β , $P_{adaptive}$ tended to increase with the number of monitoring zones as shown in Fig. 5. The value of $P_{adaptive}$ was observed to decrease with an increase in α and β . For instance, when the number of monitoring zones was 1, the values of $P_{adaptive}$ were approximately 0.55 and 0.20 with α and β set at 2 and 10, respectively. These results indicate that increasing α and β leads to better energy savings, which is a logical outcome. It turns out that the case where the number of zones changes in a situation where the number of groups does not change is not very meaningful in finding the optimal interval, δ , to minimize the energy consumption.

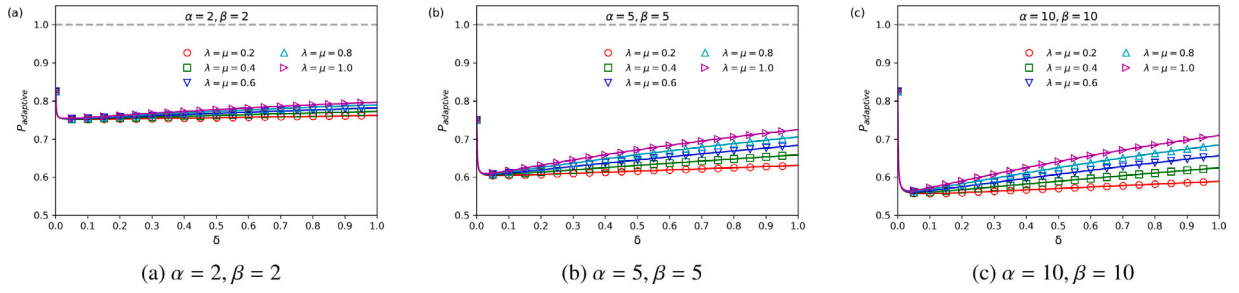
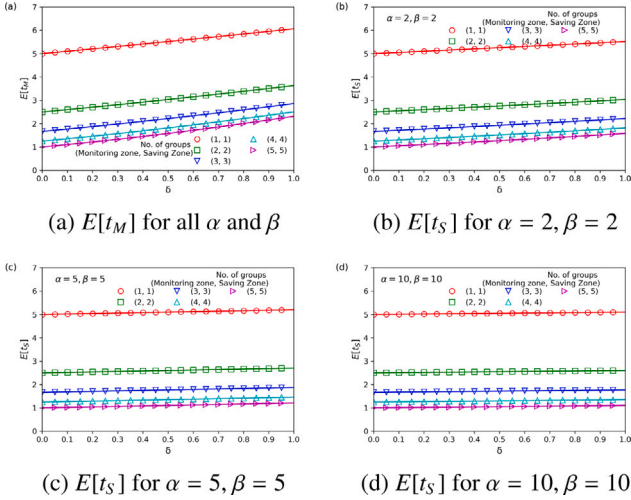
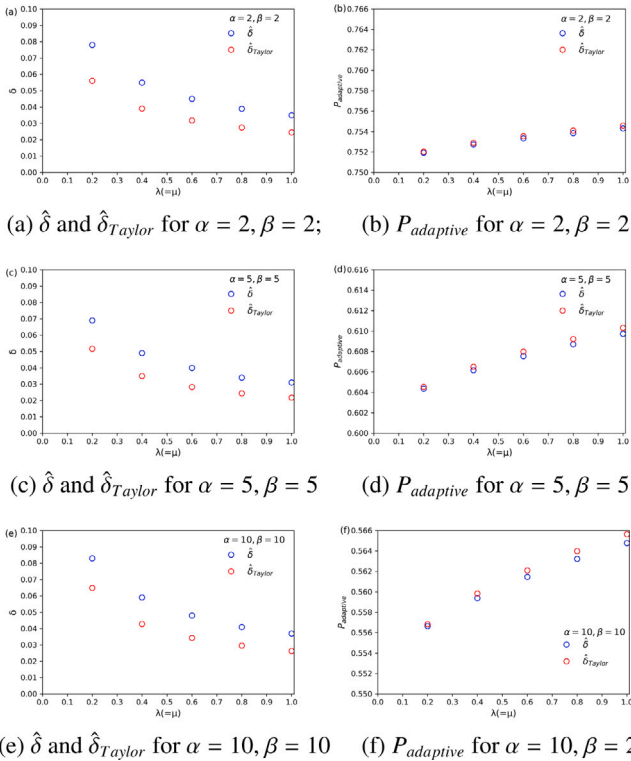


Fig. 7. Investigated cases for the evaluation of number of groups of monitoring and saving zones.

Fig. 8. Expected time in monitoring and saving zones with various λ and μ .Fig. 9. Investigated cases of various λ and μ .

4.2. Various numbers of groups of monitoring and saving zones

The proposed algorithm was investigated when the number of groups of the monitoring zones ranged from 1 to 5, and the number of groups of the saving zones remained the same as the number of groups of the monitoring zones. In this study, the number of zones in each case was set 5, and λ and μ could be calculated by dividing the number of groups by 5. As a result, the values of λ and μ were equal in each case. Fig. 6 shows the investigated cases with λ and μ including 0.2, 0.4, 0.6, 0.8, and 1.0.

As shown in Fig. 7, the value of $P_{adaptive}$ was found to be the maximum when the sampling interval was close to 0 where the investigated δ ranged from 0 to 1, and then rapidly declined before rising again as the interval increased. The inclination of the rising portion of the curve increases with the number of groups, that is, increases with λ and μ . As demonstrated in Fig. 2, an increase in δ leads to a rise in the false negative detection time in the saving zone, resulting in a longer time to recognize it as a monitoring zone. This resulted in an increase in $E[t_M]$ as shown in Fig. 8(a), which in turn led to an increase in P_M and $P_{adaptive}$. As described in Eq. (7), $E[t_M]$ is not influenced by either α or β . As explained in Eq. (9), $E[t_S]$ is significantly influenced by β and their relationship is shown in Fig. 8(b)–(d). It can be seen that the larger the β , the flatter the $E[t_S]$ curve becomes. As shown in Fig. 2, the number of false positive detections in the savings area decreases with increasing β , resulting in a reduced $E[t_S]$. The large increase in $E[t_M]$ and the relatively small increase in $E[t_S]$ resulted in V-shaped curves of $P_{adaptive}$ being consistently observed as the number of groups changed, and the optimal sampling interval to minimize $P_{adaptive}$ can be found.

In Section 3.3, we discuss that finding the optimal interval by differentiating Eq. (10) is too complex to solve. An alternative method using a Taylor series expansion as presented in Eq. (13) can be used instead. The differences between the two methods were evaluated as shown in Fig. 9, and the differences between $\hat{\delta}$ and $\hat{\delta}_{Taylor}$ were found to be less than 0.03 s. The differences in $P_{adaptive}$ values were less than 0.001 for the investigated pairs of α and β . Furthermore, P_M was set to 1 in this study; therefore, a difference of 0.001 is extremely small.

4.3. Evaluation in ground settlement cases

The proposed algorithm was used to simulate practical scenarios that may occur. In this study, we simulated two ground settlement scenarios: sinkholes in areas with limestone formations and ground settlement in areas where old sewer lines were buried [34,35]. Based on the map that contains this information (Fig. 10(a) and (b)), the zones that can be set for monitoring and saving to detect the ground settlement can be determined. The limestone area and the area where the sewer line passes have a high possibility of ground settlement; thus, they were set as monitoring zones, and the other areas were set as saving zones (Fig. 10(c) and (d)). The number of monitoring zones for the limestone area and sewer lines was 50 and 40 among 100 zones, respectively. Depending on the UGV path, the number of groups of monitoring and saving zones can vary. Therefore, the investigation was

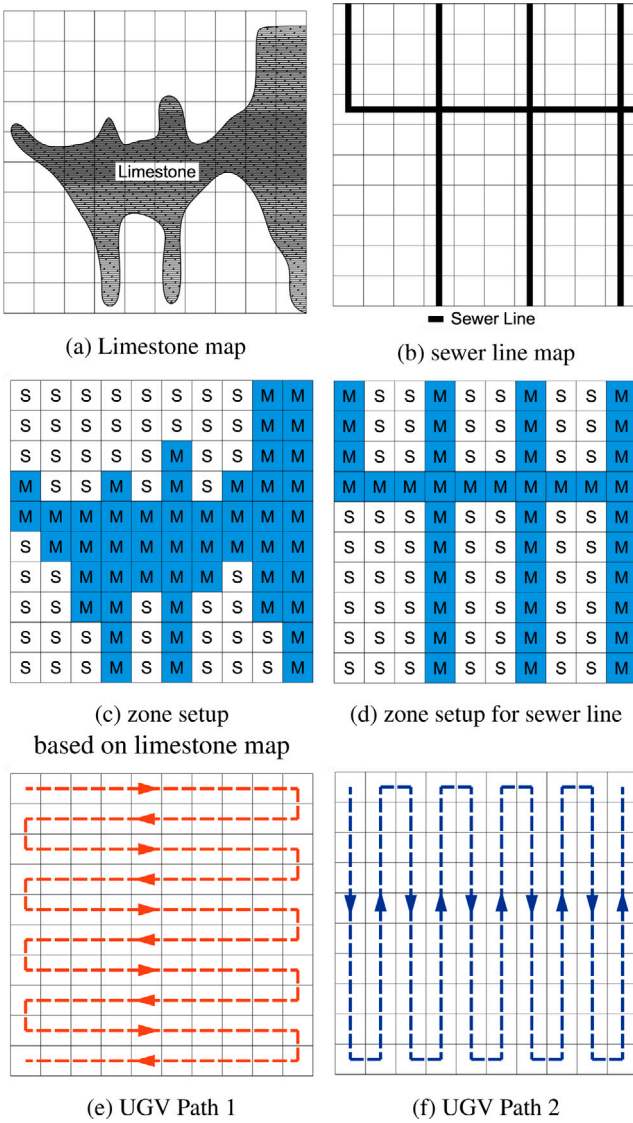


Fig. 10. Two ground settlement scenarios.

performed on Paths 1 and 2, as shown in Fig. 10(e) and (f), respectively. As previously explained, depending upon the number of those groups, the two parameters of λ and μ change even at a given number of zones; the parameters α and β were 10 in these simulation cases.

Fig. 11(a) and (b) shows the simulation results for the limestone area. The energy consumption curves differ depending on the UGV paths. The UGV consumed more energy when it followed Path 1 than when it followed Path 2. The monitoring zone for limestone was divided into 15 groups for Path 1 and 10 groups for Path 2. The corresponding values of λ and μ for Path 1 were 0.3 and 0.32, respectively, whereas for Path 2, they were both 0.2. Because Path 1 had a greater number of monitoring groups than Path 2, the number of false detections that occurred when switching between groups increased, resulting in increased energy consumption.

The simulation results for the sewer line case are shown in Fig. 11(c) and (d). Similar to the limestone case, group switching occurs more frequently in Path 1 than in Path 2. The monitoring zone was divided into 25 groups for Path 1 and 10 groups for Path 2, with corresponding values of λ and μ being 0.625 and 0.417 for Path 1, and 0.25 and 0.15 for Path 2, respectively. Because Path 1 had a larger number of monitoring groups, the frequency of false detections when switching between groups increased, leading to higher energy consumption.

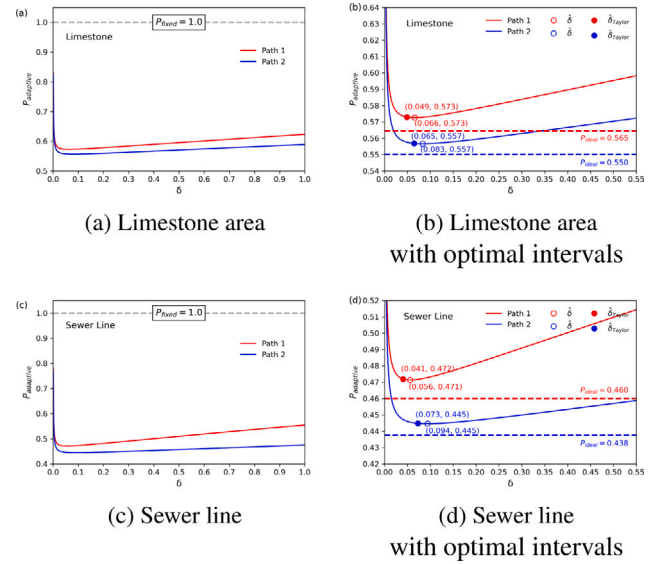


Fig. 11. Energy consumption according to path.

The optimal detection interval $\hat{\delta}$ is also plotted in Fig. 11(b) and (d). It can be seen that the optimal detection interval of Path 1 is smaller than that of Path 2. As Path 1 switches more between the monitoring and saving groups than Path 2, it is reasonable that the optimal detection interval for Path 1 is smaller than that for Path 2. The approximated optimal detection interval using the Taylor series, $\hat{\delta}_{Taylor}$, is also displayed in Fig. 11(b) and (d). It was found that the approximated interval was smaller than the ideal solution, but the energy consumption difference at these two intervals was only slight.

When monitoring the entire area without dividing it into monitoring and saving zones, P_{fixed} , which is the upper bound of power consumption, is 1. The lower bound of the power consumption is P_{ideal} plotted in Fig. 11(b) and (d). As shown in Fig. 11, the scenario of monitoring by dividing the area into zones saves energy, and the power consumption at the optimal detection interval was close to P_{ideal} .

In this study, Paths 1 and 2 were set and compared. Even under the same conditions in the monitoring and saving zones, it was found that energy consumption can vary depending on the UGV path. If various paths are used, an optimal path that minimizes the energy consumption can be found.

5. Conclusions and future works

In this study, we propose an adaptive sensor-management strategy to address the critical issue of energy consumption in UGVs by focusing on the high energy consumption of the monitoring and navigation sensors used in monitoring missions. Our study presents a novel approach for UGV monitoring analysis based on risk maps. We characterized UGV monitoring based on a risk map and derived an expected power consumption model to analyze energy efficiency. In addition, we mathematically derived the optimal sampling interval that minimizes energy consumption based on the distribution of monitoring and saving zones on the risk map. Extensive evaluations were conducted to confirm the relevance of the proposed model. We observed that adjusting the interval of the monitoring and positioning sensors according to the distribution of the monitoring and saving zones has a significant effect on reducing energy consumption. Based on the experimental results from the two ground settlement scenarios, it was observed that different tracking methods for the monitoring areas led to varying energy consumption levels, even when these areas had the same distribution.

In our future work, we plan to use a platform to consider more sensors and risk levels on the risk map to extend our model. A method

to determine the optimal path in terms of energy efficiency will also be investigated. The proposed approach is expected to be highly applicable to autonomous vehicle research, in which energy efficiency is critical.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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References

- [1] S. Bonadies, A. Lefcourt, S.A. Gadsden, A survey of unmanned ground vehicles with applications to agricultural and environmental sensing, in: *Autonomous Air and Ground Sensing Systems for Agricultural Optimization and Phenotyping*, Vol. 9866, SPIE, 2016, pp. 142–155.
- [2] O. Liu, S. Yuan, Z. Li, A survey on sensor technologies for unmanned ground vehicles, in: *2020 3rd International Conference on Unmanned Systems (ICUS)*, Harbin, China, 2020, pp. 638–645, <http://dx.doi.org/10.1109/ICUS50048.2020.9274845>.
- [3] H. Zhang, Y. Zhang, C. Liu, Z. Zhang, Energy efficient path planning for autonomous ground vehicles with ackermann steering, *Robot. Auton. Syst.* 162 (2023) 104366, <http://dx.doi.org/10.1016/j.robot.2023.104366>.
- [4] A. Sadrpour, J. Jin, A.G. Ulsoy, Mission energy prediction for unmanned ground vehicles, in: *2012 IEEE International Conference on Robotics and Automation*, Saint Paul, MN, USA, 2012, pp. 2229–2234, <http://dx.doi.org/10.1109/ICRA.2012.6224860>.
- [5] S. Amir, J. Jin, A.G. Ulsoy, Mission energy prediction for unmanned ground vehicles using real-time measurements and prior knowledge, *J. Field Robotics* 30 (3) (2013) 399–414.
- [6] J.A. Broderick, D.M. Tilbury, E.M. Atkins, Optimal coverage trajectories for a UGV with tradeoffs for energy and time, *Auton. Robots* 36 (2014) 257–271.
- [7] E. Mejri, S. Kelouwani, Y. Dube, O. Trigui, K. Agbossou, Energy efficient path planning for low speed autonomous electric vehicle, in: *2017 IEEE Vehicle Power and Propulsion Conference (VPPC)*, Belfort, France, 2017, pp. 1–6, <http://dx.doi.org/10.1109/VPPC.2017.8330908>.
- [8] E. Våljaots, R. Sell, Energy efficiency profiles for unmanned ground vehicles, *Proc. Est. Acad. Sci.* 68 (1) (2019) 55–65.
- [9] L. Hou, F. Zhou, K. Kim, L. Zhang, Practical model for energy consumption analysis of omnidirectional mobile robot, *Sensors* 21 (5) (2021) 1800, <http://dx.doi.org/10.3390/s21051800>.
- [10] J. Pentzer, S. Brennan, K. Reichard, On-line estimation of power model parameters for skid-steer robots with applications in mission energy use prediction, *J. Field Robotics* 39 (2022) 763–782, <http://dx.doi.org/10.1002/rob.22076>.
- [11] J. Bradley, E. Atkins, Optimization and control of cyber-physical vehicle systems, *Sensors* 15 (9) (2015) 23020–23049, <http://dx.doi.org/10.3390/s150923020>.
- [12] J.A. Baxter, D.A. Merced, D.J. Costinett, L.M. Tolbert, B. Ozpineci, Review of electrical architectures and power requirements for automated vehicles, in: *2018 IEEE Transportation Electrification Conference and Expo (ITEC)*, Long Beach, CA, USA, 2018, pp. 944–949, <http://dx.doi.org/10.1109/ITEC.2018.8449961>.
- [13] S. Suherman, R. Ananda, M. Pinem, Environmental sensor assessment for a light quadcopter drone, in: *IOP Conference Series: Materials Science and Engineering*, Vol. 851, IOP Publishing, 2020, (1).
- [14] A.V. Malawade, T. Mortlock, M.A. Al Faruque, EcoFusion: energy-aware adaptive sensor fusion for efficient autonomous vehicle perception, in: *Proceedings of the 59th ACM/IEEE Design Automation Conference (DAC '22)*, Association for Computing Machinery, New York, NY, USA, 2022, pp. 481–486, <http://dx.doi.org/10.1145/3489517.3530489>.
- [15] Y.E. Aslan, I. Korpeoglu, Ö. Ulusoy, A framework for use of wireless sensor networks in forest fire detection and monitoring, *Comput. Environ. Urban Syst.* 36 (6) (2012) 614–625, <http://dx.doi.org/10.1016/j.compenvurbsys.2012.03.002>.
- [16] J. Lee, A Dynamic Duty Cycling MAC Protocol for Low Latency in Low-Power and Lossy Networks (Dissertation of master's), Dept. Computer Software & Engineering, Univ. of Science & Technology, 2013.
- [17] T. Kim, H. Min, J. Jung, A mobility-aware adaptive duty cycling mechanism for tracking objects during tunnel excavation, *Sensors* 17 (3) (2017) 435, <http://dx.doi.org/10.3390/s17030435>.
- [18] S. Lee, D. Lee, P. Choi, D. Park, Accuracy-power controllable LiDAR sensor system with 3D object recognition for autonomous vehicle, *Sensors* 20 (19) (2020) 5706, <http://dx.doi.org/10.3390/s20195706>.
- [19] V. Gokhale, G.M. Barrera, R. Venkatesha Prasad, FEEL: Fast, energy-efficient localization for autonomous indoor vehicles, in: *ICC 2021 - IEEE International Conference on Communications*, Montreal, QC, Canada, 2021, pp. 1–6, <http://dx.doi.org/10.1109/ICC42927.2021.9500500>.
- [20] D. Liu, S. Wang, Z. Wen, L. Cheng, M. Wen, Y.-C. Wu, Edge learning with unmanned ground vehicle: Joint path, energy, and sample size planning, *IEEE Internet Things J.* 8 (4) (2021) 2959–2975, <http://dx.doi.org/10.1109/JIOT.2020.3023000>.
- [21] S. Wang, M. Xia, Y.-C. Wu, Backscatter data collection with unmanned ground vehicle: Mobility management and power allocation, *IEEE Trans. Wireless Commun.* 18 (4) (2019) 2314–2328, <http://dx.doi.org/10.1109/TWC.2019.2902557>.
- [22] B. Zhang, A. Davoodi, Y.H. Hu, Exploring energy and accuracy tradeoff in structure simplification of trained deep neural networks, *IEEE J. Emerg. Sel. Top. Circuits Syst.* 8 (4) (2018) 836–848, <http://dx.doi.org/10.1109/JETCAS.2018.2833383>.
- [23] H. Tann, S. Hashemi, R.I. Bahar, S. Reda, Runtime configurable deep neural networks for energy-accuracy trade-off, in: *Proceedings of the Eleventh IEEE/ACM/IFIP International Conference on Hardware/Software Codesign and System Synthesis (CODES '16)*, Association for Computing Machinery, New York, NY, USA, 2016, pp. 1–10, <http://dx.doi.org/10.1145/2968456.2968458>, Article No. 34.
- [24] D. Balemans, W. Casteels, S. Vanneste, J. Hoog, S. Mercelis, P. Hellinckx, Resource efficient sensor fusion by knowledge-based network pruning, *Internet Things* 11 (2020) 100231.
- [25] J.D. Choi, M.Y. Kim, A sensor fusion system with thermal infrared camera and LiDAR for autonomous vehicles and deep learning based object detection, *ICT Express* (2022) <http://dx.doi.org/10.1016/j.icte.2021.12.016>.
- [26] O. Elmakis, T. Shaked, B. Fishbain, A. Degani, BREEZE—Boundary red emission zone estimation using unmanned aerial vehicles, *Sensors* 22 (14) (2022) 5460, <http://dx.doi.org/10.3390/s22145460>.
- [27] S. Kilic, O. Omer, A self-adaptive UAV routing for forest fire risk mitigation: a conceptual model, in: *Proceedings of the 2019 Summer Simulation Conference*, 2019, pp. 1–12.
- [28] X. Guo, C. Yongxing, Z. Jian, H. Yuanxian, L. Bijun, HDM-RRT: A fast HD-map-guided motion planning algorithm for autonomous driving in the campus environment, *Remote Sens.* 15 (2) (2023) 487, <http://dx.doi.org/10.3390/rs15020487>.
- [29] P. Stankiewicz, T.T. Tan, M. Kobilarov, Adaptive sampling with an autonomous underwater vehicle in static marine environments, *J. Field Robot.* 38 (4) (2021) 572–597, <http://dx.doi.org/10.1002/rob.22005>.
- [30] G. Guglieri, A. Lombardi, G. Ristorto, Operation oriented path planning strategies for rpas, *Am. J. Sci. Technol.* 2 (6) (2015) 1–8.
- [31] S. Primatesta, G. Guglieri, A. Rizzo, A risk-aware path planning method for unmanned aerial vehicles, in: *2018 International Conference on Unmanned Aircraft Systems (ICUAS)*, Dallas, TX, USA, 2018, pp. 905–913, <http://dx.doi.org/10.1109/ICUAS.2018.8453354>.
- [32] K. Yao, X. Li, Uncertain alternating renewal process and its application, *IEEE Trans. Fuzzy Syst.* 20 (6) (2012) 1154–1160, <http://dx.doi.org/10.1109/TFUZZ.2012.2194152>.
- [33] Rabia Tugce Yazicigil, *Compressive Sampling As an Enabling Solution for Energy-Efficient and Rapid Wideband RF Spectrum Sensing in Emerging Cognitive Radio Systems*, Columbia University, 2016.
- [34] H.I. Mohamed, S.H. Rabey, M.S. Darweesh, Laboratory investigation of ground surface settlement caused by erosion around a leaking pipe, *J. Pipeline Syst. Eng. Pract.* 13 (1) (2022) 04021080.
- [35] T. Karoui, S.-Y. Jeong, Y.-H. Jeong, D.-S. Kim, Experimental study of ground subsidence mechanism caused by sewer pipe cracks, *Appl. Sci.* 8 (5) (2018) 679, <http://dx.doi.org/10.3390/app8050679>.



Seoyeon Kim received M.S. degree from the Department of Unmanned System Engineering at Hannam University in the Republic of Korea in 2018, and she received her Ph.D. degree from the Department of Information and Communication Engineering, Hannam University in 2022. She is currently working as a postdoctoral researcher at the Institute of Human-Centered Computing at Inha University in Incheon, Republic of Korea. Her current research interests include intelligent Internet of Things, mobile systems, edge computing, and cloud computing.



Young-Hoon Jung is a professor of civil engineering at Kyung Hee University. He had served as a vice dean at the college of engineering, KHU (2018–2021). Young-Hoon Jung received his bachelor's degree in civil engineering from Seoul National University in 1997 and master's degree in 1999. In 2005, he received his Ph.D. from Seoul National University. After doing research as a post-doc fellow at Northwestern University (2005–2007), he was hired as a senior researcher at Korea Institute of Civil Engineering and Building Technology in 2008. In 2009, he started his academic career at Kyung Hee University, where he is currently a professor. He is currently an editor-in-chief of International Journal of Geo-engineering and senior editor of KSCE Journal of Civil Engineering. His research interests are numerical and experimental modeling of fundamental soil behavior including soil arching effect and its application to geosynthetic-reinforced piled embankment system. He was a principal investigator of the national project on the smart observational method for the detection of problematic features in open excavation field in urban area.



Hong Min received his B.S. degree in Computer Science from Handong University, Pohang, Republic of Korea, in 2004 and Ph.D. degrees (Integrated Master & Ph.D. Program) in Computer Engineering from Seoul National University, Seoul, Republic of Korea in 2011. He was a visiting Postdoctoral fellow at Northwestern University, USA, after getting Ph.D. degree. He had been with the Division of Computer Information Engineering, Hoseo University, Asan, Republic of Korea, from 2013 to 2020. He has been with the School of Computing, Gachon University,



Seongnam, Republic of Korea, since 2021, where currently he is an associate professor. His research interests include embedded systems, operating systems, sensor networks, and the Internet of Things.

Taesik Kim received his B.S. degree from the Department of Civil Engineering at Seoul National University in 2000, and he received his M.S. degree from the same university in 2002. In 2011, he received his Ph.D. degree from the Department of Civil and Environment engineering at Northwestern University, IL, US. He is an associate professor at the Department of Civil Engineering, Hongik University since Sept. 2013. His current research interests include adaptive ground deformation management using IoT, landslide and drone.



Jinman Jung received his B.S. degree from the Department of Computer Science and Engineering at Seoul National University in 2008, and he received his Ph.D. degree from the Department of Electrical Engineering and Computer Science at Seoul National University in 2014. Dr. Jung was a visiting researcher at Seoul National University Institute of Computer Technology, Seoul, from Mar. 2014 to Sept. 2014. He was an assistant professor at department of Information and Communication Engineering, Hannam University since from Sept. 2014 to Feb. 2021. He is currently an associate professor in the Department of Computer Engineering at Inha University in Incheon since Mar. 2021. His current research interests include system software, distributed computing, cloud computing, and edge computing.